MIL-C-14982(MU)
15 February 1969
SUPERSEDING
PA-PD-2276

MILITARY SPECIFICATION

CABLES FOR TRAINING ATOMIC DEMOLITION
CHARGE, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers general special requirements not included on the drawings, and provide the general quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for training cables and designated as Cables for Training Atomic Demolition Charge, General Specification for.

2. APPLICABLE DOCUMENTS

2.1 Individual cables. See detail specifications (see 6.2) for applicable documents relating to individual cable assembly.

2.2 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8800545 - Bag, Packing, Ammunition, for Cable Assembly
8800546 - Bag, Packing, Ammunition, for Cable Assembly

FSC 1195

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8851893 - Box, Set-up, Paperboard, for Cable Assembly W3AP-1 (TNG)
8851896 - Box, Set-up for W41-1(A) DM (TNG) Cable Assembly
8851900 - Box, Set-up Paperboard for W41-AP-1(TNG) Cable Assembly
8851926 - Box, Set-up Paperboard for W4-AP-1(TNG) Cable Assembly
8851929 - Box, Set-up Paperboard for W4-1(A) DM (TNG) Cable Assembly
8851942 - Box, Set-up Paperboard for T4132-1 (TWG) Cable Assembly
8851943 - Box, Set-up Paperboard for T-4140 (TNG) Cable Assembly Special Purpose Branched
8851950 - Box, Set-up Paperboard for W143-6 (TNG) Cable Assembly
8851951 - Box, Set-up Paperboard for W143-7 (TNG) Cable Assembly
8851953 - Box, Set-up Paperboard for W143-9 (TNG) Cable Assembly
8851954 - Box, Set-up Paperboard for W143-10 (TNG) Cable Assembly
8852275 - Box, Set-up Paperboard for W143-11 (TNG) and W143-12 (TNG) Cable Assembly
9206885 - Box, Set-up for Cable Assembly W143-14 Training
9206886 - Bag, for Cable Assembly W143-13 Training

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail requirements. The individual requirements shall be as specified herein and in accordance with the applicable detail specifications. In the event of any conflict between requirements of this specification and the detail specifications, the latter shall govern (see 6.1 and 6.2).

3.2 Materials. The contractor shall furnish objective evidence (see 6.3) that the material complies with the specifications listed on the applicable drawing.

3.3 Assembly. The cable assembly shall comply with all requirements specified on applicable drawing (dwg.) in the detail specification (see 6.2) with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

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3.4 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.5 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.6 Circuit resistance. The cable assembly shall have the resistance specified in the applicable table in the detail specification (see 6.2) when tested as specified in 4.4.1.

3.7 Insulation resistance. The insulation resistance of any circuit specified in the detailed specification (see 6.2) shall not be less than 20 megohms when subjected to a potential of 500 ± 50 volts direct current (vdc) with the current limited to 5 milliamperes (ma) when tested as specified in 4.4.2.

3.8 Vibration. The cable assembly shall withstand vibration when tested as specified in 4.4.3 without evidence of cracks, chips, looseness of parts, distortion or other deterioration. After the test, the cable assembly shall comply with 3.6 and 3.7.

3.9 High temperature. The cable assembly shall withstand a temperature of 125 ± 5 degrees Fahrenheit (^{o}F) without evidence of swelling, peeling, cracking, distortion or other deterioration and shall comply with 3.6 and 3.7 when tested as specified in 4.4.4.

3.10 Low temperature. The cable assembly shall withstand a temperature of -13 ± 5°F without evidence of shrinkage, embrittlement, cracking, chipping, distortion or other deterioration and shall comply with 3.6 and 3.7 when tested as specified in 4.4.5.

3.11 Humidity. The cable assembly shall withstand a minimum relative humidity of 90 percent without evidence of swelling, discoloration, peeling, cracking, distortion, shrinkage, corrosion, exposure of base material or other deterioration and shall comply with 3.6 and 3.7 when tested as specified in 4.4.6.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristics.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.12.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see drawing in detail specification (see 6.2) and 3 sub-assemblies specified in detail specification (see 6.2) and 3 sets of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests need not be conducted in order specified in table I.

Table I

First article inspections and tests

	Examination or test	Contractor's Inspection before submission	Non-Destructive	Parts	Sub-Assemblies	Assemblies
1.	Classification of Characteristics (see 4.3.2.1)	X	X	X	X	X
2.	Circuit resistance (see 3.6)	X	X			X
3.	Insulation resistance (see 3.7)	X	X			X
4.	Vibration (see 3.8)	X	X			X
5.	High temperature (see 3.9)	X	X			X
6.	Low temperature (see 3.10)	X	X			X
7.	Humidity (see 3.11)	X	X			X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2, shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

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4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major characteristic table II

Lot size	Sample size 100% inspection
1- 25	25
26- 30	26
31- 35	27
36- 40	28
41- 45	29
46- 50	30
51- 55	31
56- 60	32
61- 65	33
66- 70	34
71- 75	35
76- 85	36
86- 95	37
96- 110	38
111- 130	39
131- 156	40
157- 190	41
191- 240	42
241- 340	43
341- 520	44
521-1120	45
1121 and over	

Minor characteristic table III

Lot size	Sample size 100% inspection
1- 25	16
26- 35	17
36- 45	18
46- 55	19
56- 65	20
66-100	21
101-200	22
201 and over	

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as specified in the detail specification (see 6.2) and herein.

4.3.2.1.1 All cables before potting or molding.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Soldering in accordance with MIL-S-60902	Visual/Manual
102.	No part missing, damaged or improperly assembled	Visual
103.	No foreign matter present	Visual
104.	Materials (see 3.2)	Visual
Minor:		
201.	Workmanship (see 3.12)	Visual/Manual

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4.3.2.1.2 All cables after potting or molding when potting or molding is required.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Voids and air pockets not greater than 1/16 inch diameter in molding	Visual
102.	No nicks, tears, gouges	Visual
103.	Pin not broken, flattened, nicked	Visual
104.	Connector not cracked	Visual
105.	No part missing or improperly assembled	Visual
106.	Cable tight in molding, any indication of separation between cable and molding	Visual/Manual
Minor:		
201.	Workmanship (see 3.12)	Visual/Manual

4.3.2.1.3 Preparation for delivery for production cable assemblies only (see 5.1).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual
Minor:		
201.	Workmanship (see 3.12)	Visual/Manual

4.3.2.1.4 Bag, packing, ammunition for cable assembly (see dwg 8800545, 8800546 and 9206886 when specified in Section 5 of detail specification (see 6.2) for repair parts only.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Packaging materials (see 3.2)	Visual
102.	Envelope over each connector secured with tape	Visual/Manual
103.	Cable assembly coiled to approx. 10 inch diameter and secured 3 places with tape dwg 8800545 only	Visual/Manual

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104. Length of bag before packing.....*Measure
105. Width of bag before packing.....*Measure
106. Objective evidence (see 6.3) available for heat seal test in accordance with MIL-P-116.....Visual
107. No tear, split, hole or crack in bag.....Visual

*Measure with suitable scale

Minor:
201. Marking.....Visual
202. Workmanship (see 3.12).....Visual/Manual

4.3.2.1.5 Box, set-up paperboard for applicable cable assembly (see dwgs 8851893, 8851896, 8851900, 8851926, 8851929, 8851942, 8851943, 8851944, 8851950, 8851951, 8851953, 8851954, 8852275, and 9206885 when specified in detailed spec. (see 6.2) section 5.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Connectors wrapped with cushioning and secured with tape (see dwgs 8851893, 8851926, 8851943, 8851950, 8851951, 8851953, 8851954, 8852275 and 9206885).....Visual/Manual
102. Cable packed tight.....Manual
103. Box sealed properly.....Visual
104. Corners of box blunted.....Visual
105. Air exhausted from bag.....Manual
106. Proper heat seal.....Visual/Manual
107. Objective evidence (see 6.3) available for heat seal test in accordance with MIL-P-116.....Visual
108. Packaging materials (see 3.2).....Visual
109. Marking.....Visual

Minor:
201. Workmanship (see 3.12).....Visual/Manual

4.3.2.1.6 Packing and Marking for Storage and Shipments (see dwg 8808362 and section 5 of detailed specification (see 6.2).

Categories	Characteristics	Method of inspection

Critical: None defined.

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Major:

101.	Gross load weight, max	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Package material (see 3.2)	Visual

* Determine weight with a suitable scale

Minor:

201.	Marking	Visual
202.	Workmanship (see 3.12)	Visual/Manual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

	Test	Test classification
a.	Circuit resistance	Major
b.	Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance (see 3.6) and insulation resistance (see 3.7). Each cable assembly shall be tested in accordance with 4.4.1 or 4.4.2 as applicable. If any cable assembly fails to conform to the applicable requirements, the cable assembly shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation (ET) when required. ET's or EL's as applicable shall be specified in the detail specification (see 6.2). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.5 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (see 3.6). The circuit resistance shall be measured with approved resistance equipment in accordance with table in the detail specification (see 6.2). In addition, while making each circuit resistance test specified in the table, the cable shall be gripped at a point not greater than 6 inches from the face of the connector and wiggled back and forth for more than 30 degrees of arc on each side of the axis of the cable.

4.4.2 Insulation resistance (see 3.7). The insulation resistance shall be measured with approved resistance equipment complying with 3.7 and table in the detail specification (see 6.2). In addition, while making each insulation resistance test specified in the table, the cable shall be gripped at a point not greater than 6 inches from the face of the connector and wiggled back and forth for more than 30 degrees of arc on each side of the axis of the cable.

4.4.3 Vibration (see 3.8). The cable assembly shall be mounted to a fixture by attaching each cable connector without the use of tools to a mating connector mounted on the fixture. The fixture and cable assembly shall be mounted to a suitable vibration table. The cable assembly may be supported at a point not less than 10 inches from the face of the connectors to prevent the assembly from slipping during the test. The cable assembly shall be vibrated at a frequency of 5 to 500 to 5 c.p.s. With a constant double amplitude of 0.01 inch or an applied vibratory acceleration of 2 g's (in each direction) whichever is the limiting value. The rate of cycling shall be such that 15 minutes is required to proceed from 5 to 500 and back to 5 c.p.s., and the motion shall be applied for 2 hours in each plane. Resonant frequencies shall be noted and the assembly shall be vibrated at each resonant frequency noted for a period of 15 minutes. The vibrations shall be applied in the longitudinal plane and in a plane perpendicular to the longitudinal plane. After completion of test, the assembly shall comply with 3.6 and 3.7.

4.4.4 High temperature (see 3.9). The cable assembly shall be coiled so that the diameter of the coil is not more than 12 inches. The coiled cable shall then be placed in a chamber and the temperature shall be raised to a temperature of 125 $\pm$ 5°F. After stabilization (see 6.4) the assembly shall be exposed to the temperature for 4 hours. At the end of the 4 hour period, the assembly shall be removed and immediately uncoiled to its full length. The cable assembly shall then comply with 3.6 and 3.7.

4.4.5 Low temperature (see 3.10). The cable assembly shall be tested as specified in 4.4.4 except then the temperature shall be lowered to -13 $\pm$ 5°F. After stabilization (see 6.4) the assembly shall be exposed to the temperature for 4 hours. At the end of the 4 hour period the assembly shall be removed and immediately uncoiled to its full length. The cable assembly shall then comply with 3.6 and 3.7.

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4.4.6 Humidity (see 3.11). The cable assembly shall be coiled so that the diameter of the coil is not more than 12 inches in diameter. The coiled assembly shall then be placed in a humidity test chamber. Within a period of 1 hour, the test chamber temperature shall be raised to +125°F ± 5°F stabilized (see 6.4) and held for 2 hours. A relative humidity of 90 to 95 percent shall be maintained while the test chamber is +125°F ± 5°F. At the end of the 2 hour period, the chamber temperature shall be uniformly decreased during a 1 hour period, to between + 68°F, and + 95°F. During the period of decreasing temperature, the relative humidity shall be 95 + 5 percent. The cable assembly shall be subjected to ten such humidity cycles (1 hour of increasing temperature at 95 percent relative humidity, 2 hours of temperature stabilization at 90 to 95 percent relative humidity). The water used in controlling the relative humidity shall be distilled, having a pH value of between 6.5 and 7.5 + 77°F. Upon completion of the cycles, the cable assembly shall be immediately removed from the test chamber and dried by blowing dry, filtered, compressed air over the assembly for not more than 5 minutes. The cable assembly shall then comply with 3.6 and 3.7.

5.0 PREPARATION FOR DELIVERY

5.1 Production cable assemblies used in Training Atomic Demolition Charge.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The cable assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The cable assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The cable assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair cable assemblies used in Training Atomic Demolition Charge.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The cable assembly shall be preserved and packaged in accordance with drawings specified in the detail specification (see 6.2).

5.2.1.2 Level C. The cable assembly shall be preserved and packaged in accordance with drawings specified in the detail specification (see 6.2) except that the moisture barrier may be omitted.

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5.2.2 Packing.

5.2.2.1 Level A and Level C. The cable assembly shall be packed in accordance with the drawings specified in the detail specification (see 6.2).

5.2.3 Marking. Marking shall be as specified on drawings specified on drawings specified in the detail specification (see 6.2).

6.0 NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title number and date of the detail specification (see 6.2).

b. Selection of the applicable level of packaging and packing.

6.2 Individual specifications. Cable assemblies covered under this basic specification have individual specifications assigned to each cable such as MIL-C-14982(MU)/1 for W143-5 Cable Assembly. Supplement No. 1 to this specification lists the detail specifications numerically.

6.3 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.4 Temperature stabilization. The temperature of the chamber is considered "stabilized" when three consecutive temperature readings, taken at 5-minute intervals, are within $\pm$ 1°F. of each other.

6.5 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

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Preparing activity:
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MILITARY SPECIFICATION

CABLES FOR TRAINING ATOMIC DEMOLITION
CHARGE, GENERAL SPECIFICATION FOR

MIL-C-14982/1(MU) - Cable Assembly, T-4132-1; Training (see dwg 8850777)
MIL-C-14982/2(MU) - W4-1(A) DM Cable Assembly Training (see dwg 8850944)
MIL-C-14982/3(MU) - W143-6 Cable Assembly Training (see dwg 8850948)
MIL-C-14982/4(MU) - W143-7 Cable Assembly Training (see dwg 8850949)
MIL-C-14982/5(MU) - W143-9 Cable Assembly Training (see dwg 8850951)
MIL-C-14982/6(MU) - W143-10 Cable Assembly Training (see dwg 8850952)
MIL-C-14982/7(MU) - T-4140 Cable Assembly Training (see dwg 8850954)
MIL-C-14982/8(MU) - Cable Assembly XM6 Antenna (see dwg 8850956)
MIL-C-14982/9(MU) - Cable Assembly Firing Device (see dwg 8851127)
MIL-C-14982/10(MU)- W3-AP-1 Cable Assembly Training (see dwg 8851879)
MIL-C-14982/11(MU)- W4-AP-1 Cable Assembly Training (see dwg 8851880)
MIL-C-14982/12(MU)- W143-11 Cable Assembly Training (see dwg 8852184)
MIL-C-14982/13(MU)- Cable Assembly (see dwg 8852203)
MIL-C-14982/14(MU)- Cable Assembly Control Receiver (see dwg 8852509)
MIL-C-14982/15(MU)- W143-12 Cable Assembly Training (see dwg 8852528)
MIL-C-14982/16(MU)- Cable Assembly W143-14 Training (see dwg 9203770)
MIL-C-14982/17(MU)- Cable Assembly W143-13 Training (see dwg 9207513)
MIL-C-14982/18(MU)- Cable Assembly W41-AP-1E1 Training (see dwg 9216954)
MIL-C-14982/19(MU)- Cable Assembly W41-1(A)DM-E1 Training (see dwg 9216955)
MIL-C-14982/20(MU)- Cable Assembly Special Purpose Electrical T4194 TNG (see dwg 9229265)

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